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The integration of CAD/CAM and Rapid Prototyping in Product Development: A review

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Abstract

Rapid Prototyping and manufacturing (RP&M) technique has shown a high potential to reduce the cycle and cost of product development. Integrating CAD/CAM RP&M systems have been developed and employed to implement remote parts and manufacturing for rapid prototyping and improve the capability of rapid product development for a large number of small and medium sized enterprises. This paper provides a comprehensive review of recent research on CAD/CAM RP&M systems. The key issues and enabling tools to implement the remote RP&M systems, involve (1) CAD/CAM/CAE, (2) Modelling and Optimization, (3) Algorithms, (4) Measurement of anisotropic strength, (5) STEP/XML, (6) Applying new technologies and concepts to Rapid prototyping systems, are described in detail.Finally, this review gives outlook on possible future development and research direction for CAD/CAM RP&M systems.

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*Keywords:*CAD/CAM; Rapid prototyping; algorithms.

1. Main text

Due to the pressure of international competition and marketglobalization in the 21st century, there continues to be strongdriving forces in industry to compete effectively by reducing time to market and cost while assuring high quality product andservice [5][3][4].RP is a new forming process which fabricates physical parts layer by layer under computer controldirectly from 3D CAD models in a very short time. In contrast totraditional machining methods, the majority of rapid prototypingsystems tend to fabricate parts based on additive manufacturingprocess, rather than subtraction or removal of material. Therefore,this type of fabrication is unconstrained by the limitationsattributed to conventional machining approaches [2][6].Additive processes, which generate parts in a layeredway, have 15 years of history [1].It started in the late 80's with Stereo lithography. Sincethen, many new ideas have come up, many patents havebeen deposited, new processes were invented andcommercialized, some of which have alreadydisappeared. An overview is given in Table 1.

Table 1. LM. technologies, acronyms and development years

Name	Acronym	Development Years

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Stereolithography	SLA	1986-88
Solid Ground curing	SGC	1986-99
Laminated Object Manufacturing	LOM	1985-91
Fused Deposition Modeling	FDM	1988-91
Selective Laser Sintering	SLS	1987-99
3D Printing	3DP	1985-97

RP is newly evolving toward rapidtooling (RT). RT is a technique that transforms the RP patterns intofunctional parts, especially metal or plastic parts. It offers a fast andlow cost method to produce moulds and functional parts.Furthermore, the integration of both RP and RT in developmentstrategy promotes the implementation of concurrent engineeringin companies. Numerous processes have been developed forproducing molds from RP systems. The RT methods can generallybe divided into direct and indirect tooling categories, and also soft(firm) and hard tooling subgroups. Indirect RT requires some kindsof master patterns, which can be made by conventional methods(e.g. High-speed Machining, HSM), or more commonly by an RPprocess such as SL or SLS. Direct RT, as the name suggests, involvesmanufacturing a tool cavity directly on the RP system, henceeliminating the intermediate step of generating a pattern [7]. Softtooling can be obtained via replication from a positive pattern ormaster. Soft tooling is associated with low costs; used for low volumeproduction and uses materials that have low hardnesslevels such as silicones, epoxies, low melting point alloys, etc. [53] RTVsilicone rubber moulds, epoxy moulds, metal spraying moulds, etc., are some of these typical soft molds. Hard tooling is associatedwith higher volume of production, and the use of materials ofgreater hardness. Keltool process, Quickcast process, and theExpress Tool process are some of these hard tooling's. Electricaldischarge machining (EDM) seems to be an interesting area inwhich rapid tooling finds a potential application. Some methods ofmaking EDM electrodes based on RP technique have developedsuch as abrading process, copper electroforming and net shapecasting, etc. [37][38][39].

The objective of this paper is to present a comprehensive review of recent advances in CAD/CAM RP&M systems, anddiscuss the future research directions. That may provide areference and direction for the development of CAD basedRP&M systems.

2. Definition and Classification of Rapid Manufacturing

The major players in this field have already come to theconclusion that a breakthrough can only happen on thebasis of manufacturing applying the RP technologies ona large scale. New terms were created or considered:terms like mass customization (MC) by Siemens andPhonak [1] [8] [9], Production on Demand (POD) by Boeing,and recently Advanced Digital Manufacturing (AMD) by3D Systems.

RM must guarantee long-term consistentcomponent use for the entire product life cycle or for adefined minimal period for wearing parts. This calls for amost significant role of materials in the LM technologies.The indirect productionvia patterns and reproduction, by e.g. casting, are notconsidered as LM, even though generative processeshave a significant ability and importance representing animportant option in those process chains.In contrast to the previous definition of RapidManufacturing (RM), Rapid Tooling (RT) aims only atlong-term consistency tools, meaning a tool able to formseveral thousands or even millions of parts before finalwearing-out.
Referring to tooling, mainly plastic injection moulds areconsidered, as these are the most frequently usedforming tools. Naturally we have to consider as well otherneeds with greater difficulty as in die casting, sheet metalforming and forging dies [10][11][12][13] [14]. Thoseapplications increase substantially the requirements onthermal and mechanical load and wear. On the otherhand we have also to consider tooling with lower loadthat is also of great interest: for example forthermoforming, fibre forming and similar processes [15][16].Fig. 2shows a flowchart ofdeveloping and manufacturing process for RT basedon the integrated system[22].

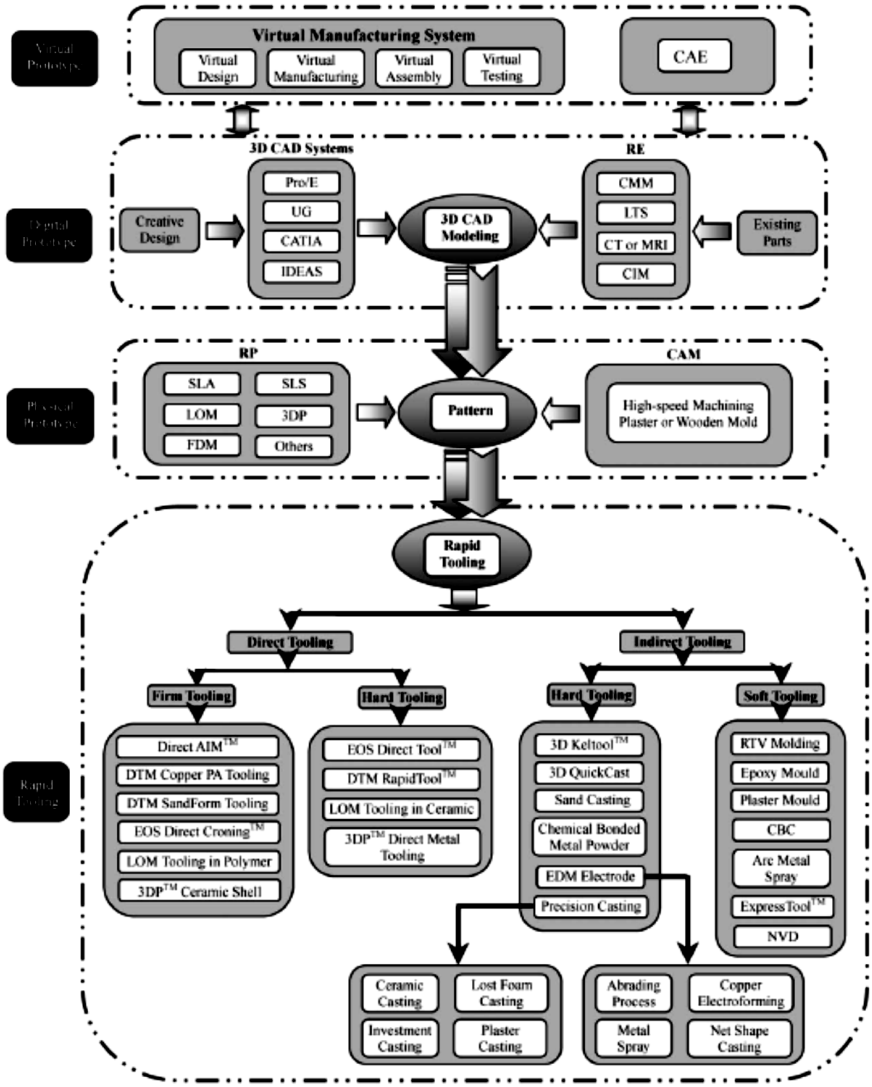

Fig. 1.Architecture of the RT integrated manufacturing system.

3. Key issues in product Development

In the development of various types of CAD RP&Msystems, a number of key issues have been discussed. Based on thecurrent research results for CAD RP&M systems, these keyissues can be divided into the following six categories: (1) CAD/CAM/CAE, (2) Modelling and Optimization, (3) Algorithms, (4) Measurement of anisotropic strength, (5) STEP/XML, (6) Applying new technologies and concepts to Rapid prototyping systems, are described in detail.

3.1. CAD/CAM/CAE

STL, which has been a widely adopted data standard in the RP/ M industry, is an approximation representation scheme of product models based on triangles or quadrilaterals. However, there are some intrinsic problems in STL files such as gaps, holes, missing/ degenerated/overlapping facets, etc. during the conversion process from native

CAD files. [17–21]. On the other hand, STL is less inaccurate in geometrical representation and needs much more storage spaces for a complex model than parametric mathematical representation themes such as NURBS, Bezier, B-Spline, etc. Recently, there are active research and development to introduce parametric mathematical representation to better support RP/M. [23-28].
Some of the research successfully adopted parametric representation such as Bezier, B-Spline and NURBS to improve the geometrical quality of RP/M models. However, a research gap exists between the parametric representation and the following process planning, and a smart solution is imperative to address the different geometrical characteristics of complex biomedical models in order to improve the build efficiency and geometrical accuracy of RP/M.

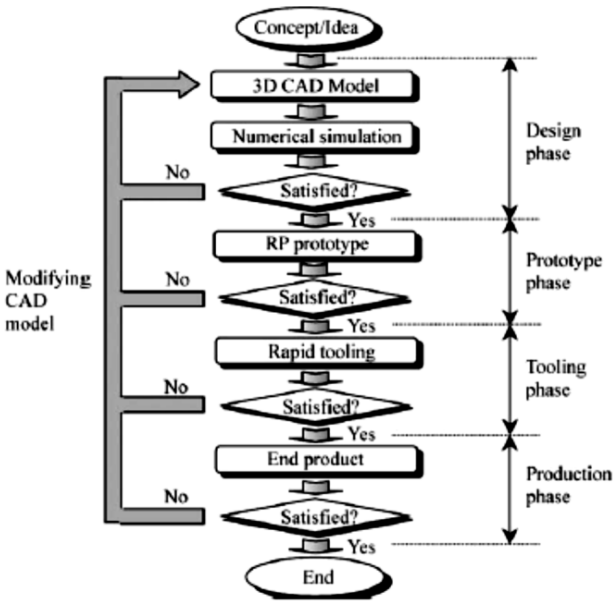

Fig. 2. Flow chart of developing and manufacturing RT based on the integrated system.

3.2. Modelling and Optimization

The modelling module allows the designer to interact with the VR model to manipulate the viewing perspective and the part geometry. This allows the designer to view the model by navigating around it and changing the lighting, shading and rendering conditions [42-50].

For example Working with the high stress had been observed in the right lift arm of thethree point hitch assembly on the 8000 Series tractor due to high vertical forces. These stresses onlyoccurred in the right lift arm because the load that caused the stresses was offset to the right of the center of the tractor. This problem was selected because the stresses could be analyzed using "finiteelement analysis and because a shape change in the area around the hub (area of high stress) couldpossibly interfere with the lift cylinder rod. The lift arms are located on the rear of the tractor. They are cast parts and are the input links ina four bar mechanism that is used to raise and lower equipment that is mounted to the hitch. Theright and left lift arms are connected to each other via the rockshaft. There are two hydrauliccylinders, one connected to each lift arm, that apply the force to raise and lower the arms. Lift linksconnect the lift arms to the draft links where the implement is mounted [51] [52].

3.3. Measurement of Anisotropic Compressive Strength

In order to make compression test specimens, an ABS material for FDM, azp102 (plaster powder) material for the 3D printer, and an acrylic-hydroxyapatite composite material for NCDS. The dimensions of the compressionspecimens are according to ASTM D695.In compression tests, build direction was the only examined processparameter. “axial” (horizontal) and “transverse”(vertical). The raster angles were setas (45◦/−45◦). Note that (45◦/−45◦) are the default raster angles defined in theQuickslice program. For the 3D printer process, the specimens were produced in three builddirections—“axial,” “transverse” and “diagonal”. The diagonal specimen’slayers were assumed to be deposited at a 45◦ slope to the base.[11] [30-33]

3.4. Algorithm

A slicing algorithm is to represent the boundary contour of each sliced layer as a closed NURBS curve to maintain the representation accuracy of an original model. A mixed tool-path algorithm is to generate contour and zigzag tool-paths to meet both the geometrical accuracy and build efficiency requirements. The contour tool- paths are used to fabricate the area along the boundary of each sliced layer to improve the geometrical quality of a product model. The zigzag tool-paths are used to fabricate the interior area of the model to improve the efficiency. [8] [29].

[6]In Adaptive slicing the first layer obtained from one end with a sufficiently small thickness. Points in this layer are projected along the RP building direction onto a plane positioned at the component edge. This forms a 2D band of scattered points. An intuitive method is to use the width of this band as a measure for adaptive slicing. Adaptation of a thick layer in components with a large curvature variation along the building direction normally results to a thick scatter band width. It is assumed that a thick band width generally represents a large cusp height and consequently a large shape-error. A band width tolerance is given to set the maximum allowable deviation between the generated model and the cloud data points. If the measured maximum band width is smaller than the prescribed band width tolerance, the layer thickness is adaptively increased. Since the actual shape-error can only be determined after a layer is constructed, the slicing procedure follows an iterative process. In this respect, construction-check- reconstruction procedure is performed until a suitable band width tolerance compatible with a satisfactory shape-error is achieved.

In DSP algorithms designed usingMATLAB (version 6) Simulink DSP block sets implemented in real-time through RTW.RTW generates C code from the graphical block sets throughthe Target Language Compiler (TLC) [14]. Together withother supporting files, the codes are compiled, assembledand linked automatically using TI TMS320C6000 codegeneration tools and GMAKE utility. The executablecode is then downloaded to the TMS320C6701-EVM forreal-time signal processing. The rapid prototyping processcan be summarized into three processes:
(1) Code generation process;
(2) Executable code building process; and
(3) Executable code downloading process.

3.5. STEP/XML

The key function in the structure of this layer is the data storage and retrieval based on STEP based data modelstandard. STEP standard has data protocols on which the structure ofproduct data-form its design information and data related to its production process data is organized. The structure thathas been used for implementing this standard was based onthe integrity of product data [40]. The STEP data model iscomposed of substructures called Application Protocols (APs). These APs include definitions not only of typicalgeometry and drafting elements, but also of data types andprocesses for specific industries such as automotive, aerospace, shipbuilding, electronics, plant construction, and maintenance [13][41]. Considering the INFELT STEP platformto support CAD/CAPP/CAM processes and CNC machiningefforts, the following Application protocols are used inIntegration layer:[34-39]
1. Configuration controlled design
2. Functional data and schematic representationfor process plans
3. Core data for automotive mechanical designprocesses
4. Mechanical product definition for processplanning
5. Material information for products
6. integrated CNC machining

7. Product life cycle support

3.6. Applying new technologies and concepts to Rapid prototyping systems

3D printer builds parts in layers. From the model data of STL, and slicing algorithm generated with detailedinformation for every layer. First, a layer of powder is spreadover the machine bed. The component slice is then formed byink jets which selectively inject a binder into the material to forma green part. A piston lowers the part, the next powder layeris spread and the binder is printed again. This layer-by-layerprocess repeats until the part is completed. Following a post process,unbound powder is removed, leaving the fabricatedpart. TheNano composite deposition system (NCDS)uses polymer resins as matrix and various nanoparticles to form composite materials, and consists of depositionmechanism and material removal process of "mechanical micromachining" .The nano composite deposited into 10–100_m thin layersto form a near-net-shape. By repeating the depositionand machining for each layer, final three-dimensional part will obtain.

4. Conclusions

CAD/CAM and Rapid prototyping have thepotential to further improve the traditional manufacturing andservice; moreover, they will become more and moreimportant for current manufacturing industry. CAD-basedRP&M systems have been developed and employed to implementremote products and manufacturing for rapid prototyping, enhancethe availability of RP&M facilities and improve the capability ofrapid product development for a large number of small andmedium sized enterprises. This paper reviews the state-of-the-artin CAD-based RP&M systems. The future improvement tasks forthe systems should further focus on the followingaspects: In a collaborative environment, the integration andinteroperability can enhance the competitive advantages of RP service. Therefore, the integration and interoperabilityin CAD-based RP&M systems is an important issue. CAD servicesrepresent a step forward in enabling collaborations betweenvarious entities on the design and in overcoming the scaling problems that may appear. CAD services technique mayprovide a solution for this issue. Some new technologies and concepts such as, STEP, XML; provide effective enabling tools for software-basedapplication systems. Future research and development on CAD-based RP&M systems should be based on these new
technologies and concepts, in particular 3Dprinting, NCDS.Further research will be focused on concurrent process for RP and RT, optimizingexternal manufacturing resources. CAD-based design and manufacturing system is a new productmode in term of mission, structure, infrastructure, capabilities, anddesign process. The significant contribution of this paper is to present acomprehensive review of recent advances in CAD-based RP&Msystems, and discuss the future research directions. That mayprovide a reference and direction for the development of CAD basedRP&M systems. AlthoughCAD-based technologies have been applied tocollaborative product design and manufacturing for many years, real industrial applications have not been in place yet. As one of themost typical cases for CAD-based collaborative product design andmanufacturing system, CAD-based RP&M system has shown a promising prospect for manufacturing. However, there is still a long way for really commercialapplication of the CAD-based RP&M systems.

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